

# Guide to Survive in EC Lab@NTOU

## EC Lab@NTOU 研究生指南

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# Outline

- 1 Research Topics & Lines of Study (研究主題與學習課程)
- 2 Seminars & Meetings (研究會議)
- 3 Tools (工具)
- 4 Duties (職責)
- 5 Mutter (嘮叨)



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# 研究主題與方向

- Randomized Algorithms (隨機演算法)
- Computational Social Choice (計算社會選擇學)
- Machine Learning Theory (機器學習理論)
- Algorithmic Game Theory & Mechanism Design (賽局理論與演算法 & 機制設計)
- Online Learning (在線學習)



# 研究生學習規劃

- ① 修畢系上要求必修科目
- ② 定期閱讀論文 (高品質之會議論文與期刊論文)
  - 參考 CS 研討會排名: [Conference Ranks 連結](#) (可以 Qualis 為準)
  - 參考 CS 期刊排名: [SJR 連結](#)
- ③ 從研讀論文與準備報告中找到有興趣的題目或方向
  - 或者由老師指定國科會計畫相關題目
- ④ 訂下論文主題或方向後，研讀相關文獻。以論文為主軸，不必廣，但求深
- ⑤ 修習或旁聽與論文相關課程，強化研究能力與技巧



# Recommended Courses for Randomized Algorithms

- Data Structures (資料結構)
- Computer Algorithms (計算機演算法)
- Probability Theory & Statistics (機率與統計)
- Discrete Mathematics (離散數學)
- Randomized Algorithms (隨機演算法)\*



# Recommended Courses for Machine Learning Theory

- Computer Algorithms (計算機演算法)
- Probability Theory & Statistics (機率與統計)
- Mathematics for Machine Learning (機器學習數學)\*
- Randomized Algorithms (隨機演算法)
- Economics and Computation (經濟與計算)\*



# Recommended Courses for Algorithmic Game Theory

- Computer Algorithms (計算機演算法)
- Probability Theory & Statistics (機率與統計)
- Mathematics for Machine Learning (機器學習數學)
- Randomized Algorithms (隨機演算法)\*
- Economics and Computation (經濟與計算)\*





# Recommended Courses for Online Learning

- Computer Algorithms (計算機演算法)
- Probability Theory & Statistics (機率與統計)
- Mathematics for Machine Learning (機器學習數學)
- Randomized Algorithms (隨機演算法)\*
- Online Learning (在線學習)\*



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# Meetings

- Weekly Group Meeting.
- Weekly Friday Seminar @IIS, Academia Sinica (PI: Prof. Chi-Jen Lu)
- **Theory Day** in Taiwan.
- Possible joint meetings with other professors.
  - Prof. Chih-Chieh Hung @IIS, NCHU
  - Prof. Po-An Chen @IIS, NYCU
  - Prof. Chi-Jen Lu@IIS, Academia Sinica



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# Coding & Writing

- C/C++, Python (Tensorflow/ PyTorch, Scikit-learn), ...
- $\text{\LaTeX}$  for writing reports and making slides (Beamer).
  - Can be done @Overleaf website.
  - A [reference project](#) as a template.
- NTOU Thesis [template](#).



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# Must Have before Graduation (畢業必備要項)

- Get your paper accepted in CMCT (滿足 NTOU CSE 申請口試規定)
- Thesis proposal (論文計畫書)
- $\geq 5$  times acknowledged paper presentations (五次以上由老師認可之論文報告)
  - 以實驗室五人輪值為例，一年半 (78 週; 最後半年忙碩士論文) 共有：  
(78 週 - 期中與期末考週  $\times 4$  - 寒暑假休假 4 週 - 其他休假 5 週)/5 人 = 13 週次/人。
- Thesis oral examination (論文口試)

## 其他研究生職責

- 擔任助教 (資料結構; 程式語言; 線性代數; 機器學習數學, etc)
- 研究生相關庶務 (報帳, 請購, ...)
- 蒐集國科會計畫相關文獻
- 管理伺服器
- 實驗室整潔
- 研討會籌辦事務

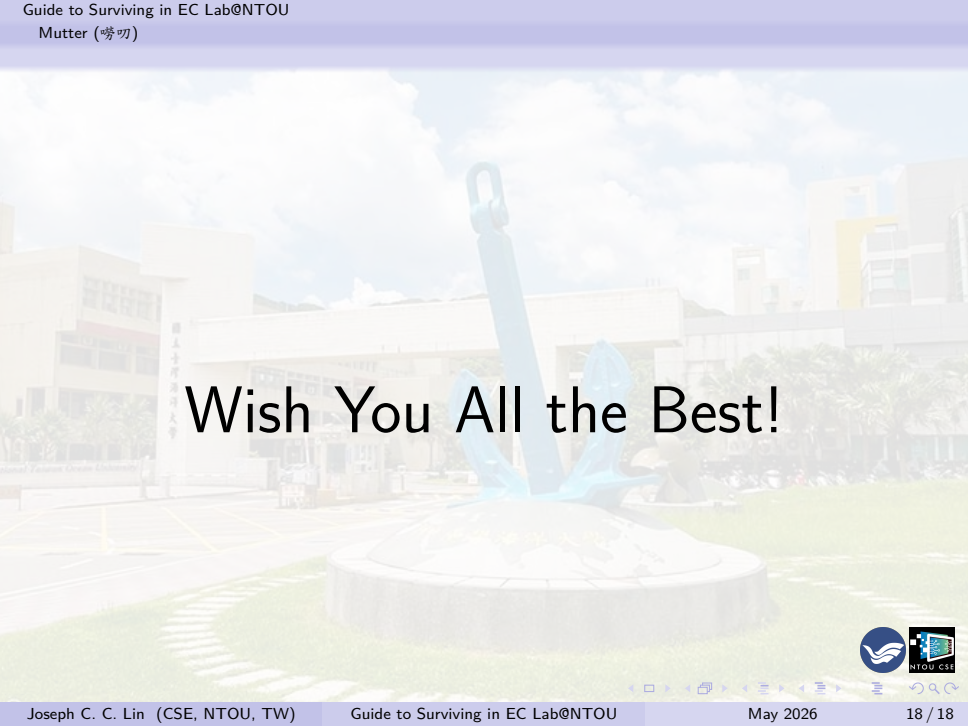




## 一些嘮叨與忠告

- 態度決定一切；請保持積極的態度
- 件件工作反映自我，凡經我手必為佳作
- 追求卓越，不要妄自菲薄
- 畢業的時間掌握在自己手裡，請做好自己的時間規劃





# Wish You All the Best!

